

On a traditional futures exchange, if the price drops by 5 percent investors are required to replenish the collateral to bring it back up to 10 percent. If investors fail to do this, the exchange liquidates the position. A similar mechanism exists on dYdX, but with important differences. First, if any position falls to 5 percent, keepers will trigger liquidation. If any collateral remains, they may keep it as a reward. Second, the liquidation is almost instantaneous. Third, no centralized exchange exists. Fourth, dYdX contracts are perpetual, whereas traditional exchange contracts usually have a fixed maturity date.²⁹

Consider the following example. Suppose the BTC price index is 10,000 USDC/BTC. An investor initiates a long position by depositing 1,000 USDC as margin (collateral), creating a levered bet on the price of BTC. If the price rises by 5 percent, the profit is 500. Given the investor has only deposited 1,000, the investor's rate of return is 50 percent, or $(1,000 - 500)/1,000$.

We can also think about the mechanics another way. Taking a long position at 10,000, the investor is committing to buying at 10,000 and the obligation is 10,000. Think of the obligation as a negative balance because the investor must pay 10,000 according to the contract. The investor has already committed collateral of 1,000 and owes 9,000. On the other side, the investor has committed those funds to purchase an asset, 1 BTC. The investor thus has a positive balance